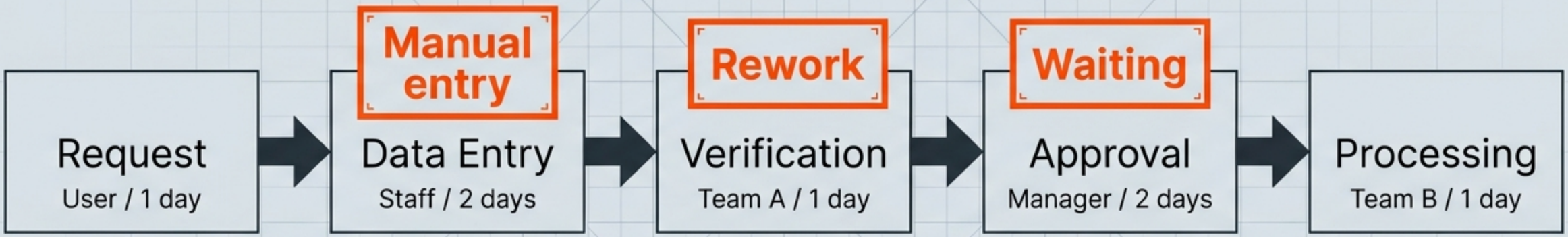
Engineering Sustainable Perfection

Process Improvement & DMAIC Final Report

Project Lead:	Jane Doe
Department:	Global Operations
Project Sponsor:	Michael Chen
Master Black Belt Assessor:	Dr. Sarah Jenkins
Date:	December 15, 2026

Gemini Notebook

Value-added steps are currently buried under waiting loops and manual rework



Initial observation revealed a disjointed workflow where transit and queue times severely outpace actual processing.

Gemini Notebook

Process inefficiencies create severe ripple effects across customers and the bottom line

10 Days
Average Lead Time (vs 5 expected)

18% Rework Rate

Baseline Period:
January – June 2026



Tollgate 1
Checked:
DEFINE
phase complete

Gemini Notebook

Targeting a 50% reduction in lead time with strictly defined operational boundaries



Lead Time:
Baseline 10 days
→ Target 5 days



Rework Rate:
Baseline 18%
→ Target <5%

Timeline: June to December 2026

IN Scope

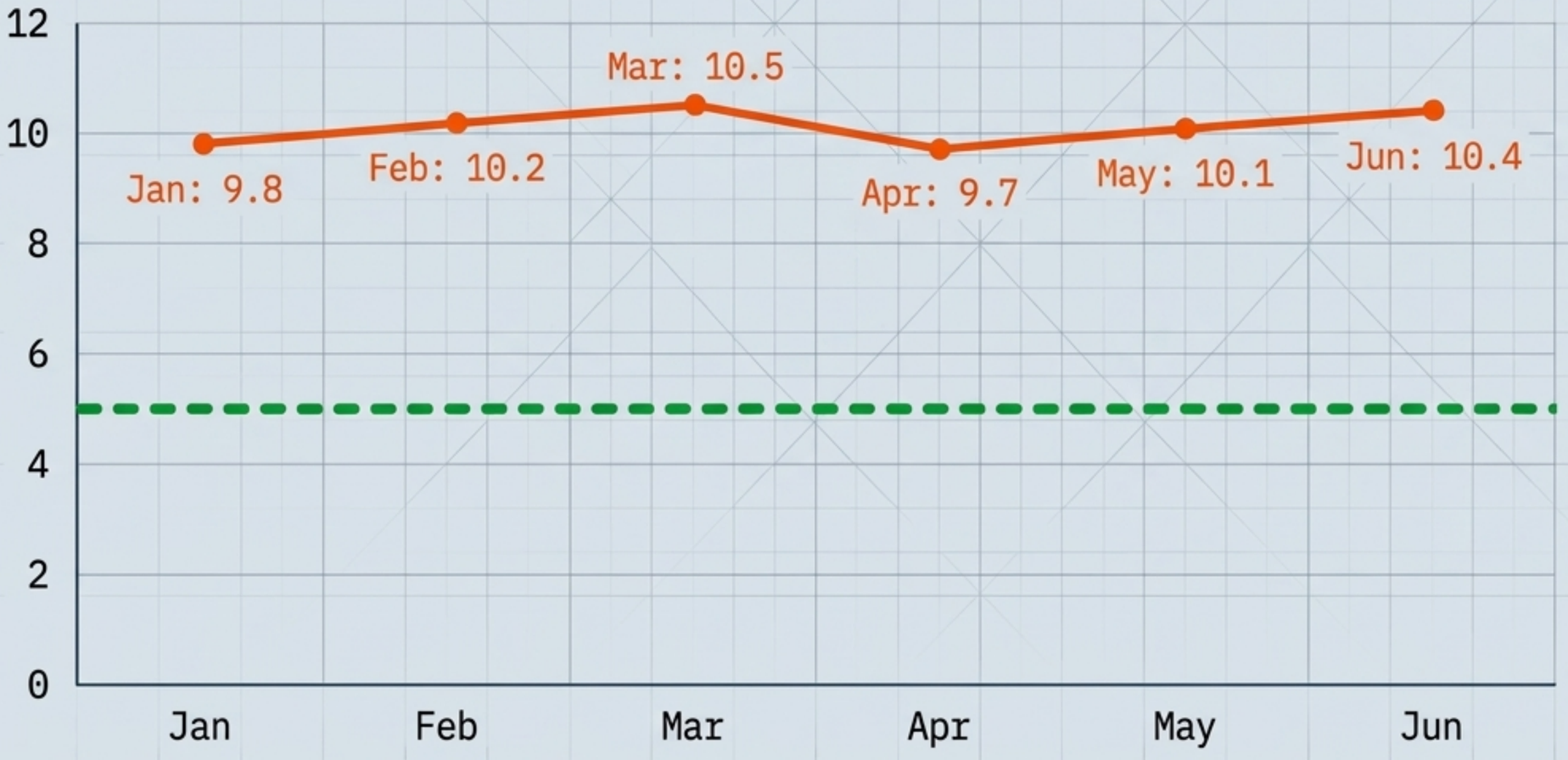
Process A, Departments A & B, Activities strictly from request generation to final completion.

OUT of Scope

Customer onboarding, legacy processes outside the immediate workflow, other operational departments.

Gemini Notebook

Six months of baseline data demonstrate a systemic failure rather than seasonal variance

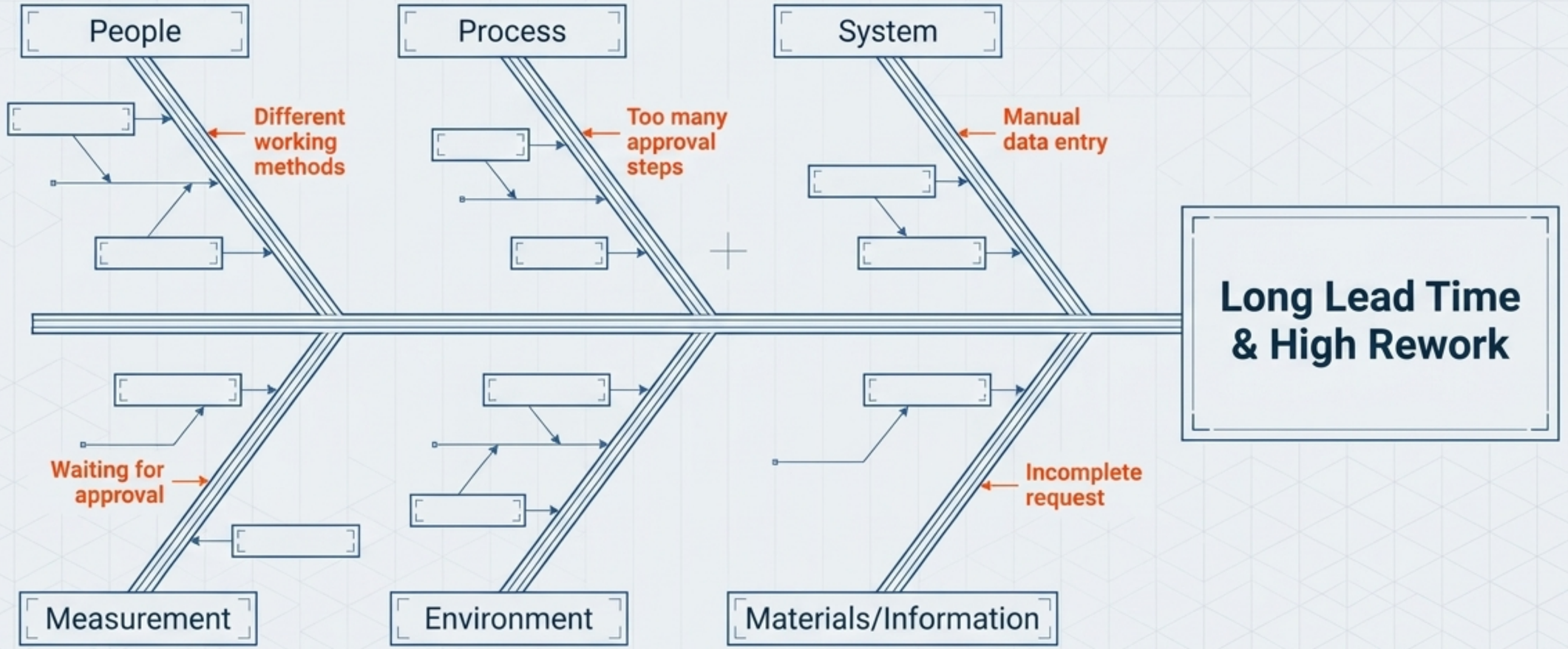


Data Sources
Validated via System Logs, Excel Extracts, and Database Records.

Tollgate 2
Checked:
MEASURE
phase complete

Gemini Notebook

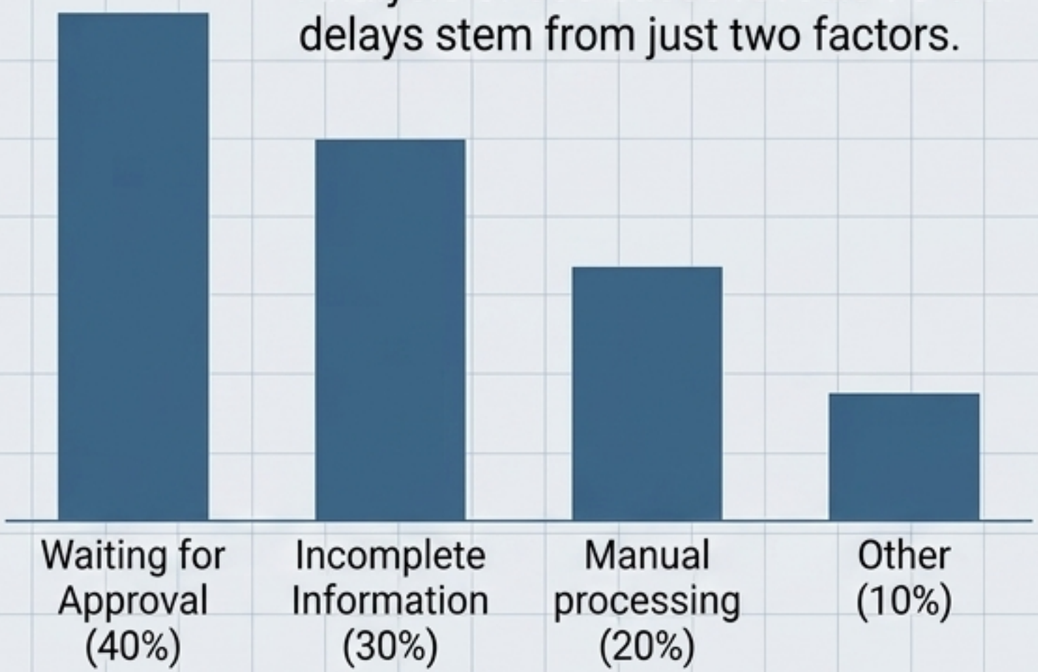
Mapping the structural and environmental universe of potential process failures



Gemini Notebook

Separating initial operational assumptions from data-validated root causes

Analysis of 500 cases reveals 70% of delays stem from just two factors.



Potential Cause	Data Evidence	Result
Staff Workload	Low Correlation	[X] Discarded
System Performance	No Significant Impact	[X] Discarded
Incomplete Information	Strong Correlation	[V] Validated
Approval Waiting	Strong Correlation	[V] Validated

Tollgate 3 Checked: ANALYZE phase complete

Gemini Notebook

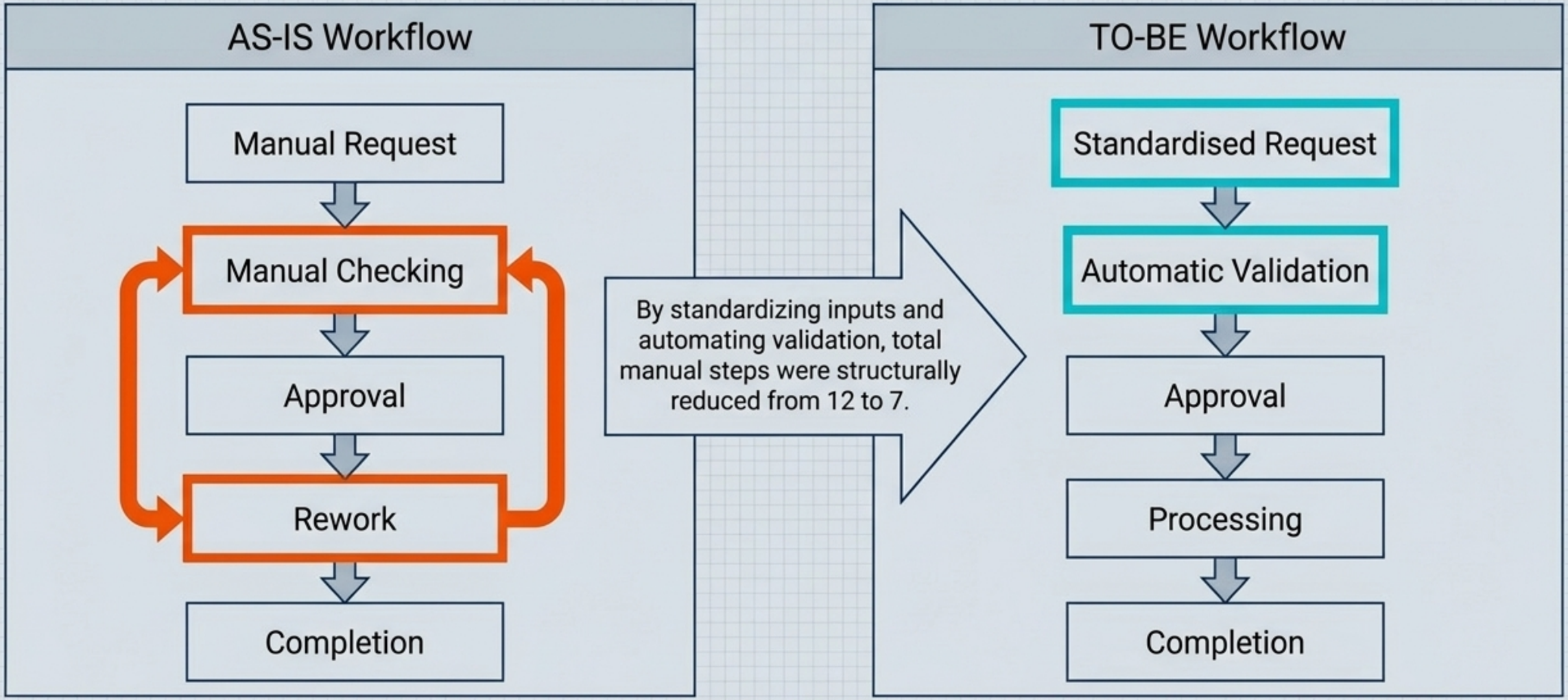
Evaluating potential interventions based on impact, cost, and implementation difficulty

Solution Evaluation Matrix				
Intervention	Impact	Cost	Difficulty	Final Decision
Additional Staff				✕ Rejected
Automation				V Selected
New Approval Workflow				V Selected
Training Only				✕ Rejected

Automation and Workflow redesign provide the optimal sustainable ROI, directly targeting the validated root causes without prohibitive recurring personnel costs.

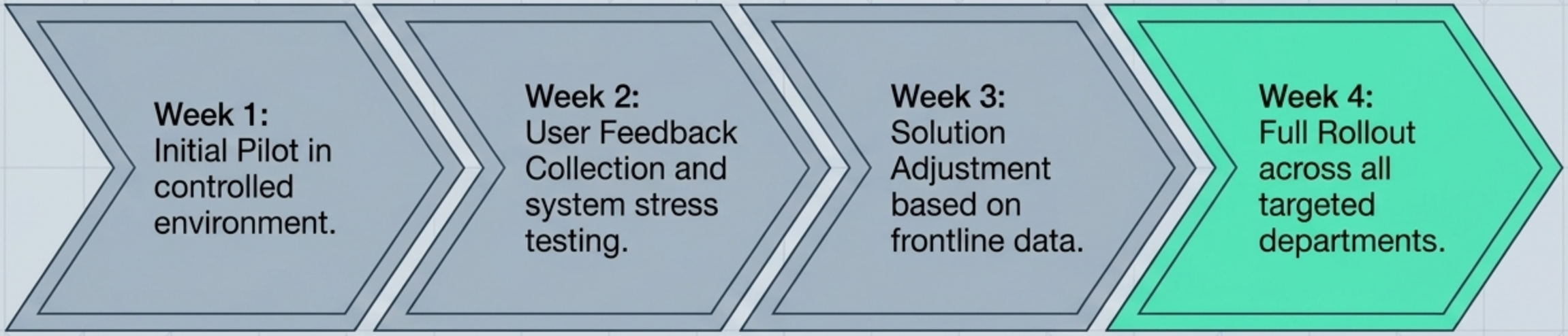
Gemini Notebook

Structurally eliminating manual waste and forced forced rework loops



Gemini Notebook

Mitigating deployment risk through a controlled, four-week iterative rollout



Testing in a controlled environment secured crucial user buy-in and refined the mechanics prior to enterprise deployment.

Tollgate 4
Checked: IMPROVE phase complete

Gemini Notebook

Process transformation yields drastic reductions in lead time and manual effort

Lead Time: 10 days → 5 days	50% ↓
Rework Rate: 18% → 4%	78% ↓
Manual Steps: 12 → 7	42% ↓



The streamlined workflow reclaimed 167 operational hours per month, drastically reducing customer wait times and internal error rates.

Gemini Notebook

Engineering a resilient ecosystem to sustain operational gains

Control Plan			
Metric	**Target**	**Frequency**	**Owner**
Lead Time	≤ 5 days	Weekly	Process Owner
Rework	< 5%	Monthly	Team Lead
Defect Rate	< 2%	Monthly	QA



KPI Dashboard Created



SOPs Updated

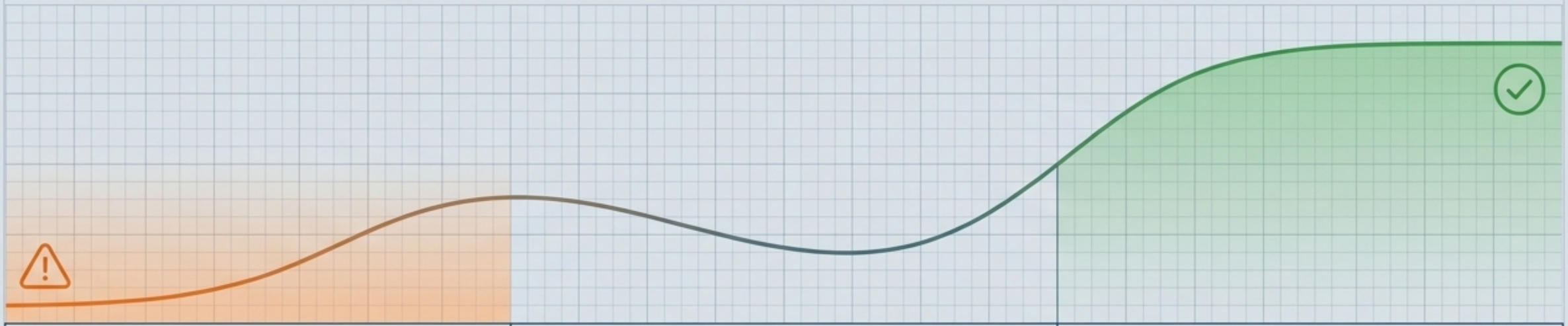


Automated Alerts Activated

Tollgate 5
Checked: CONTROL phase complete

Gemini Notebook

Technical solutions fail without psychological ownership and change management



Before Change	During Change	After Change
Existing habits, initial resistance, fear of additional work.	Transparent communication, intensive training, pilot feedback loops.	Continuous monitoring, active support, long-term standardization.

"Different stakeholders have different priorities. Early involvement creates psychological ownership, proving that process improvement relies heavily on human dynamics, not just tools."

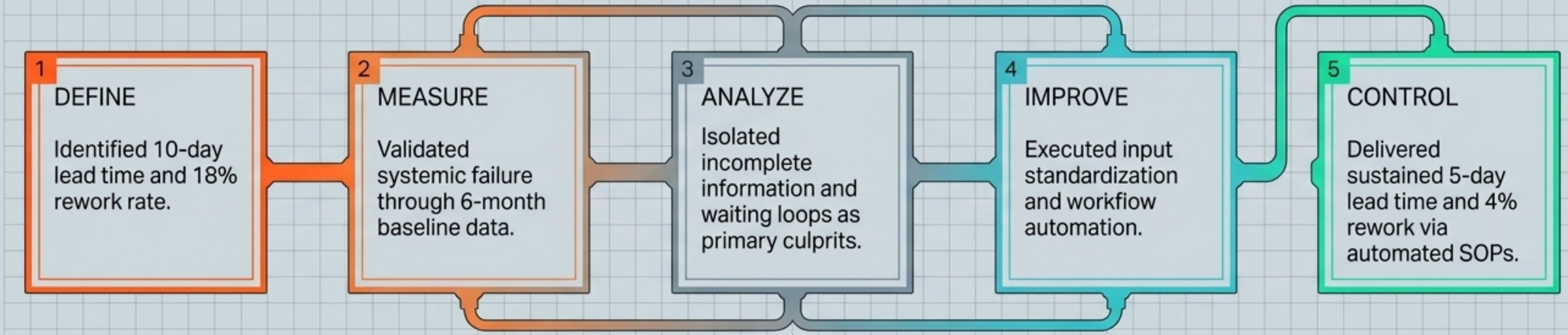
Gemini Notebook

Continuous improvement requires critical self-reflection on the DMAIC execution

Focus Area	Execution Reality	Future Adjustments
Data Collection	Collected baseline successfully	Initiate data collection earlier in the Define phase.
Stakeholders	Held regular status meetings	Involve frontline end-users much earlier in the design process.
Solution Design	Implemented a successful pilot	Test a wider array of technical alternatives before selection.
Control Phase	Built a manual KPI dashboard	Fully automate monitoring to remove reporting friction.

Gemini Notebook

The transformation from baseline chaos to sustainable, engineered control



"The objective was not simply to make the process faster, but to eliminate root causes and engineer sustainable perfection."

Gemini Notebook